TRANSFORMING THE IT DEPARTMENT IN THE DX/DC ERA

FROM "SYSTEM BUILDERS" TO "DC CYCLE CATALYSTS"

PART I:

THE EVOLUTION OF IT ROLES

Polarization into "Vanishing Functions" and "Enhanced Functions" as business departments internalize digital skills.

I-1 VANISHING ROLES: OC SUPPLY

- Requirement Definition (Order-taking style)
- Data Extraction & Simple Reporting
- SaaS & Low-code Deployment Support
- Standard Systemization Requests

↓ DECREASING VALUE: ORDINARY CAPABILITY SUPPLY

The Core Problem

"These are merely the supply of Ordinary Capabilities (OC). As business units become autonomous, the demand for these centralized functions disappears."

I-2 ENHANCED ROLES (1): ARCHITECTURE & R&D

Architecture Manager

Administering Tech Architecture

Managing foundations that cannot be built by individual units: GxP Cloud, Security, API Standards, and Global Authentication.

Edge Tech Explorer

Technology R&D

Proactive testing and evaluation of frontier technologies: Quantum Computing, Next-gen AI agents, and Synthetic Data.

I-2 ENHANCED ROLES (2): ECOSYSTEM & CATALYSIS

Ecosystem Integrator

Connecting Diverse Actors

Designing data standardizations and API hubs connecting CROs, CMOs, and Tech Partners as "Co-specialized Assets."

DC Cycle Catalyst

The Organizational Neuro-system

Linking digital signals across business units to elevate them into organizational decisions. Facilitating the DC cycle.

CORE MECHANISM: THE DC CYCLE

01 Sensing: Cross-departmental antenna for tech intelligence and regulatory shifts.

02 Seizing: Providing sandboxes and decision-support dashboards for rapid PoCs.

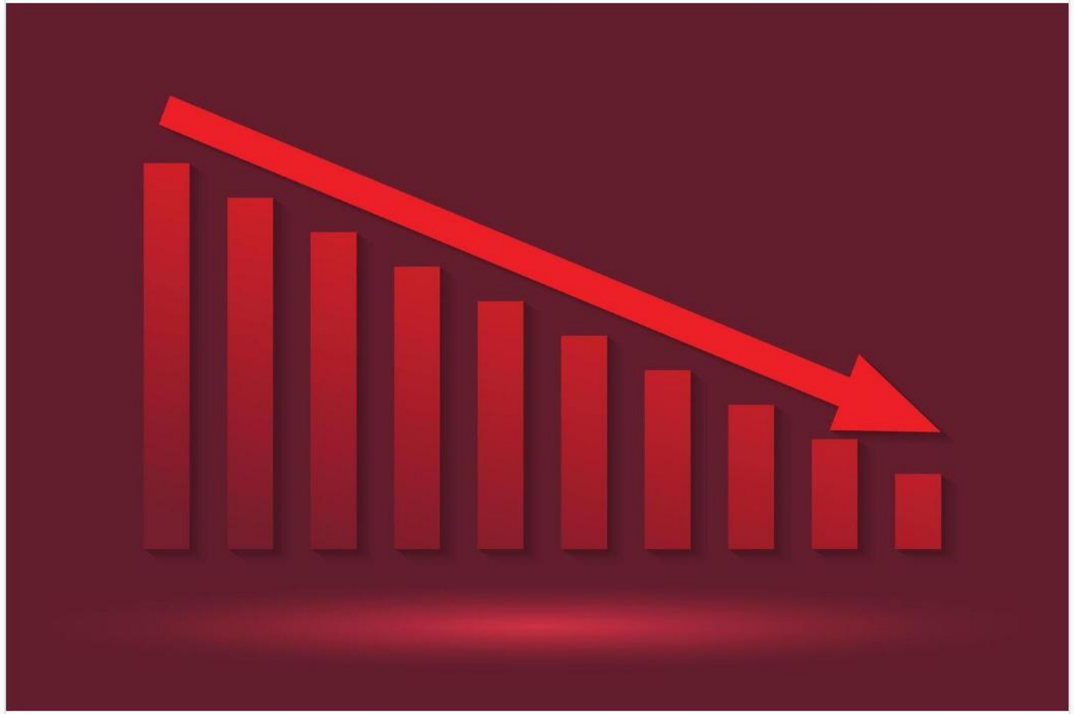
03 Transforming: Orchestrating cross-silo data integration and cultivating digital culture (CoE).

II THREE-LAYER CAPABILITY FRAMEWORK

Technical Capability	**The Foundation:** Enterprise Arch, Data Mesh, Zero Trust Security, GxP Validation.
Business Bridging	**The Translator:** Value Chain Understanding, Product Management, Data Literacy.
DC Orchestration	**The Catalyst:** Facilitating Sensing, Seizing, and Transforming support functions.

TECHNICAL CAPABILITY: HIGH-END EXPERTISE

Exclusive technical domains that remain centralized despite decentralized digital utilization.



Key Focus Areas

- **EA Design:** Multi-cloud integration & Microservices.
- **Data/AI Infra:** Data Fabric & MLOps framework.
- **Compliance:** 21 CFR Part 11 & GxP-validated environments.

BUSINESS BRIDGING: THE TRANSLATOR ROLE

Domain Expertise

Deep understanding of the Pharma Value Chain: R&D, Clinical, Mfg, and PV.

Product Management

Shifting from "Project Completion" to "Continuous Value Delivery" using OKRs.

Insights Conversion

Converting sensing signals into business insights via Data Stewardship.

3-1 SKILL REQUIREMENT MATRIX

ROLE	CORE SKILLS	NEW ESSENTIAL SKILLS
EA	Cloud, APIs, Security	Ecosystem Design, GxP Cloud, AI Infra
Data/AI Engineer	Data Pipelines, ML Infra	MLOps, AI Governance, RWD Integration
Product Manager	Agile, Scrums	Domain Deepening, DC Sensing, OKRs
Business Technologist	Project Management	Pharma Value Chain, Data Translation

SUMMARY: THE NEW MISSION

"The value of IT lies not in **WHAT** is built, but in functioning as the **NEURO-SYSTEM** that enables the organization to sense, seize, and transform."

Q&A

Thank you for your attention.

IMAGE SOURCES

https://static.vecteezy.com/system/resources/previews/068/443/186/non_2x/minimalist-red-downward-arrow-graph-decline-bar-chart-illustration-free-vector.jpg

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